GOLDEN GATE UNIVERSITY

SAN FRANCISCO

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DEPARTMENT: MASTER OF BUSINESS ADMINISTRATION

MSBA 324: Data and Social Media Analytics

Assignment 4: - R and Big Data

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Setup and Data Loading: -

Setup: -

Code:

install.packages("tidyverse")

install.packages("janitor")

library(tidyverse) library(janitor)

screenshot: -

```
> install.packages("tidyverse")
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/tidyverse_2.0.0.tgz'
Content type 'application/x-gzip' length 428840 bytes (418 KB)
==================================================
downloaded 418 KB


The downloaded binary packages are in
	/var/folders/6m/m6zfmnsd6_v4kdzc4j8cf8_w0000gn/T//RtmpZErIMS/downloaded_packages
> install.packages("janitor")
trying URL 'https://cloud.r-project.org/bin/macosx/big-sur-arm64/contrib/4.5/janitor_2.2.1.tgz'
Content type 'application/x-gzip' length 287795 bytes (281 KB)
==================================================
downloaded 281 KB


The downloaded binary packages are in
	/var/folders/6m/m6zfmnsd6_v4kdzc4j8cf8_w0000gn/T//RtmpZErIMS/downloaded_packages
> library(tidyverse)
── Attaching core tidyverse packages ──────────────────────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
> library(janitor)

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test

>
```

Data Loading: -

Code:

Load the data

Set the full path to my cleaned file

file_path<- "/Users/ronithmantheni/Documents/python_files/cleaned_airport_data.csv"
faa_data <- read_csv(file_path)

Clean column names

# This turns "Airport ID" into "airport_id", which is much easier to work with in R
faa_data <- faa_data %>% clean_names()

screenshot: -

```
> # Load the data
> # Set the full path to your cleaned file
> file_path <- "/Users/ronithmantheni/Documents/python_files/cleaned_airport_data.csv"
> faa_data <- read_csv(file_path)
Rows: 33800 Columns: 29
── Column specification ────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
Delimiter: ","
chr  (17): Airport ID, Operator Type, Operator Subtype, Aircraft Class, Aircraft Type, Damage, Effect, FAA Region, Operator, Phase Of Flt, Precip, Remarks,...
dbl  (11): Aircraft Mass, Aircraft Out of Service, Birds Seen, Birds Struck, Cost Other Infl Adj, Cost Repairs Infl Adj, FLT, Index Nr, Nr Injuries, Nr Eng...
date  (1): Incident Date

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
>
> # Clean column names
> # This turns "Airport ID" into "airport_id", which is much easier to work with in R
> faa_data <- faa_data %>% clean_names()
> |
```

Code:

Check the data (optional)

glimpse(faa_data)

screenshot: -

```
> # 3. Check the data (optional, but good practice)
> glimpse(faa_data)
Rows: 33,800
Columns: 29
$ airport_id              <chr> "KROC", "KRSW", "KOAK", "KGRI", "KCVG", "KBHM", "KCLT", "KDEN", "KPVD", "KDTW", "KPVD", "KVIS", "KHVN", "KSMF", "KIWA", "KSMF…
$ operator_type           <chr> "Private and Business", "Domestic Passenger", "Domestic Passenger", "Private and Business", "Domestic Passenger", "Domestic P…
$ operator_subtype        <chr> "Business", "Domestic Carriers", "Domestic Carriers", "Business", "Codeshare Carriers", "Domestic Carriers", "Codeshare Carri…
$ aircraft_class          <chr> "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airplane", "Airp…
$ aircraft_mass           <dbl> 2250.0, 149500.5, 149500.5, 16350.5, 16350.5, 149500.5, 149500.5, 149500.5, 149500.5, 149500.5, 149500.5, 16350.5, 16350.5, 1…
$ aircraft_out_of_service <dbl> 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 4, 0, 0, 0, 0, 0, 0, 0, 0, 8, 0, 0,…
$ aircraft_type           <chr> "C-172", "A-320", "B-737-700", "LEARJET-45", "EMB-145", "MD-88", "CRJ700", "B-737-700", "A-320", "EMB-170", "EMB-170", "BE-19…
$ birds_seen              <dbl> 55.5, 1.0, 1.0, 55.5, NA, NA, NA, NA, NA, NA, 1.0, NA, 1.0, NA, NA, NA, 1.0, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 55.5, NA…
$ birds_struck            <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, NA, NA, 1, 1, 1, 1, 1, 1, 1, NA, NA, 1, 1, 1, N…
$ cost_other_infl_adj     <dbl> 0, 0, 0, 0, 0, 106, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
$ cost_repairs_infl_adj   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ damage                  <chr> "Damaged to unknown extent", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown…
$ effect                  <chr> "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unk…
$ faa_region              <chr> "East", "South", "Western Pacific", "Central", "South", "South", "South", "Northern Mountains", "Northeast", "Great Lakes", "…
$ flt                     <dbl> NA, 465, 435, NA, 4367, 1372, 2212, 623, 474, 5814, 3338, NA, 4210, 2064, NA, 812, 1461, 4994, NA, NA, NA, NA, NA, NA, 3782, …
$ incident_date           <date> 2012-12-13, 2012-12-13, 2012-12-13, 2012-12-13, 2012-12-14, 2012-12-14, 2012-12-14, 2012-12-14, 2012-12-14, 2012-12-15, 2012…
$ index_nr                <dbl> 330507, 330508, 330509, 330510, 330511, 330513, 330515, 330516, 330517, 330519, 330520, 330522, 330523, 330525, 330526, 33052…
$ nr_injuries             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
$ nr_engs                 <dbl> 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, NA, NA, NA, NA, NA, 2, 2, 2, 2, 2, 2, 2, 2, 4, 1, 2, 2, 2, 2, 2, 2, 1, 2, …
$ operator                <chr> "BUSINESS", "JETBLUE AIRWAYS", "SOUTHWEST AIRLINES", "BUSINESS", "AMERICAN EAGLE AIRLINES", "DELTA AIR LINES", "PSA AIRLINES"…
$ phase_of_flt            <chr> "Approach", "Approach", "Climb", "Approach", "Approach", "Landing Roll", "Approach", "Approach", "Landing Roll", "Landing Rol…
$ precip                  <chr> "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown", "Unk…
$ remarks                 <chr> "OUT L/E OF LT WING DMGD DUE TO STRIKE.", "JBU REPTD STRIKING ONE LRG BIRD UPON LANDING APPROX 100 FEET ABOUVE THE T/D ZONE F…
$ size                    <chr> "Medium", "Medium", "Small", "Small", "Small", "Small", "Unknown", "Small", "Small", "Small", "Small", "Small", "Medium", "Sm…
$ sky                     <chr> "No Cloud", "Overcast", "No Cloud", "Some Cloud", "Overcast", "Some Cloud", "No Cloud", "No Cloud", "Some Cloud", "Overcast",…
$ species                 <chr> "American crow", "Northern harrier", "Unknown bird", "Perching birds (y)", "Unknown bird", "Dunlin", "Unknown bird", "Lapland…
$ speed                   <dbl> 75, 130, 155, 90, 138, NA, NA, NA, NA, NA, NA, 127, NA, 145, NA, NA, 150, NA, NA, NA, NA, NA, NA, NA, 200, 110, 120, 135, NA,…
$ state                   <chr> "NY", "FL", "CA", "NE", "KY", "AL", "NC", "CO", "RI", "MI", "RI", "CA", "CT", "CA", "AZ", "CA", "NY", "NC", "PA", "PA", "PA",…
$ time_of_day             <chr> "Day", "Day", "Night", "Day", "Day", "Day", "Dusk", "Day", "Day", "Night", "Day", "Dawn", "Night", "Day", "Night", "Day", "Da…
>
```

1. Big Data Analytics: Wildlife Strikes by Airport...

Code:

```
 # 1. Generate the frequency table
# im adding filter(!is.na(airport_id)) to safely remove any missing airport IDs first.
 airport_strikes <- faa_data %>%
filter(!is.na(airport_id)) %>%
count(airport_id, name = "strike_count", sort = TRUE)

# Get just the top 10
top_10_airports <- airport_strikes %>%
 slice_head(n = 10)

# --- DEBUGGING STEP ---
# This line will print the top_10_airports table in my console.
# we can see a table with 10 rows before the plot appears.
print(top_10_airports)
```

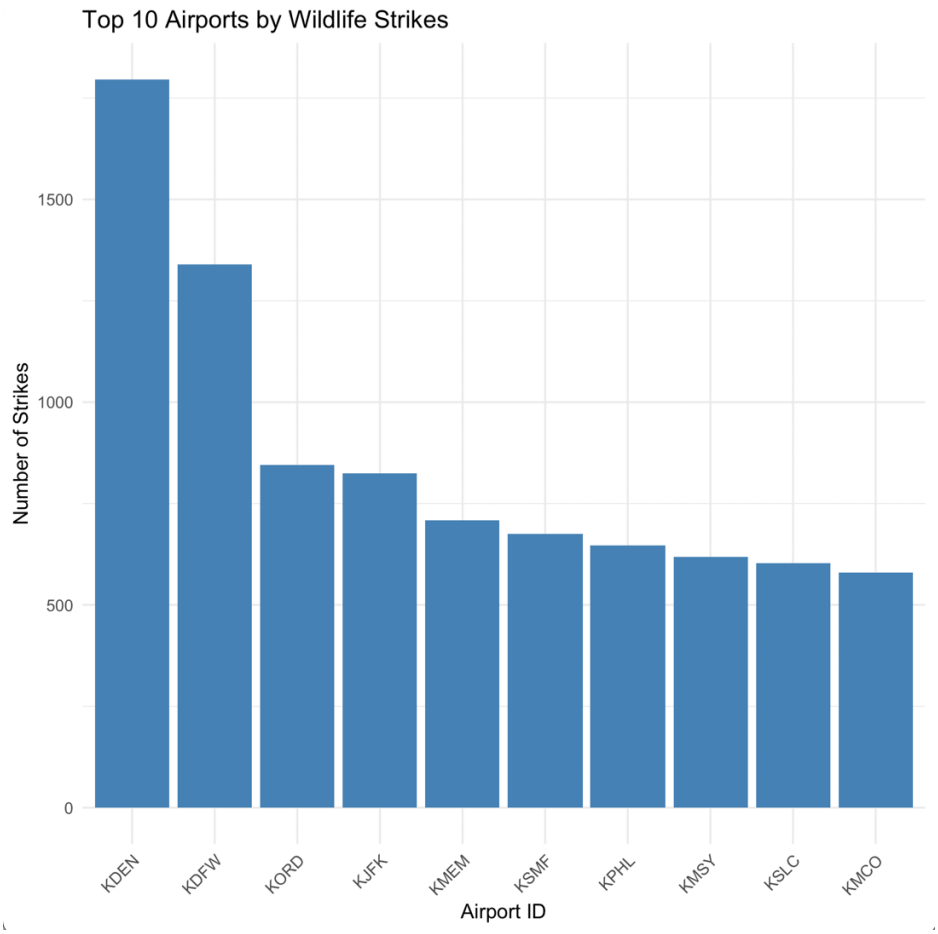
screenshot: -

```
> # 1. Generate the frequency table
> # We add filter(!is.na(airport_id)) to safely remove any missing airport IDs first.
> airport_strikes <- faa_data %>%
+   filter(!is.na(airport_id)) %>%
+   count(airport_id, name = "strike_count", sort = TRUE)
>
> # Get just the top 10
> top_10_airports <- airport_strikes %>%
+   slice_head(n = 10)
>
> # --- DEBUGGING STEP ---
> # This line will print the top_10_airports table to your console.
> # You should see a table with 10 rows before the plot appears.
> print(top_10_airports)
# A tibble: 10 × 2
   airport_id strike_count
   <chr>             <int>
 1 KDEN               1796
 2 KDFW               1341
 3 KORD                845
 4 KJFK                825
 5 KMEM                708
 6 KSMF                674
 7 KPHL                648
 8 KMSY                619
 9 KSLC                602
10 KMCO                579
>
```

bar plot code: -

```
#  Generate the bar plot

ggplot(top_10_airports, aes(x = reorder(airport_id, -strike_count), y =
strike_count)) +

    geom_bar(stat = "identity", fill = "steelblue") +

    labs(title = "Top 10 Airports by Wildlife Strikes",

                 x = "Airport ID",

                 y = "Number of Strikes") +

     theme_minimal() +

theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

screenshot: -



2.Big Data Analytics: Wildlife Strikes by Aircraft Type

Code:

```
#Generate the frequency table
# This is a more robust way to filter. It first converts the text to lowercase.
# and trims any whitespace, then it filters out "unknown".
 aircraft_strikes <- faa_data %>%
 mutate(aircraft_type = str_to_lower(aircraft_type)) %>%    # Convert to lowercase
mutate(aircraft_type = str_trim(aircraft_type)) %>%              # Trim whitespace
 filter(aircraft_type != "unknown") %>%                        # Filter out all variations
count(aircraft_type, name = "strike_count", sort = TRUE)

# Get just the top 10 known aircraft
top_10_aircraft <- aircraft_strikes %>%
 slice_head(n = 10)
# Print the table to my console
print(top_10_aircraft)
```

screenshot: -

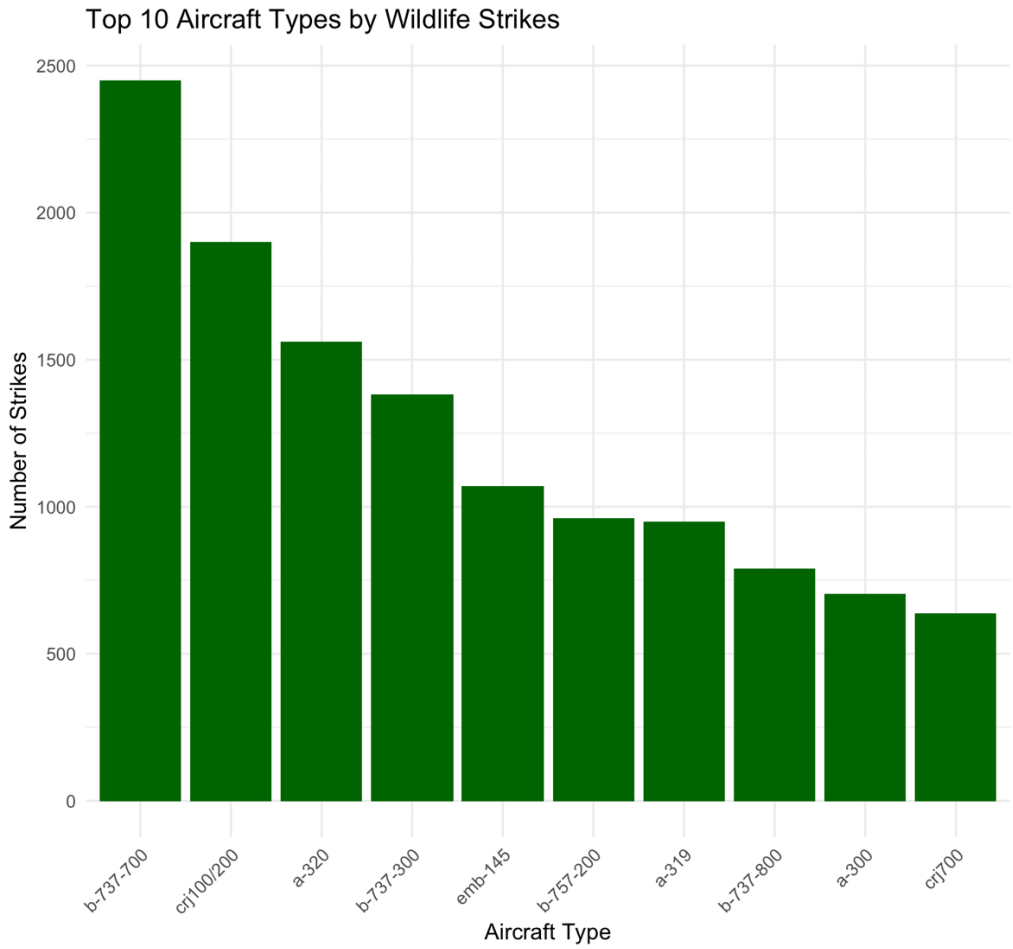
```
> # 1. Generate the frequency table
> # This is a more robust way to filter. It first converts the text to lowercase
> # and trims any whitespace, then it filters out "unknown".
> aircraft_strikes <- faa_data %>%
+   mutate(aircraft_type = str_to_lower(aircraft_type)) %>% # Convert to lowercase
+   mutate(aircraft_type = str_trim(aircraft_type)) %>%     # Trim whitespace
+   filter(aircraft_type != "unknown") %>%                  # Filter out all variations
+   count(aircraft_type, name = "strike_count", sort = TRUE)
>
> # Get just the top 10 known aircraft
> top_10_aircraft <- aircraft_strikes %>%
+   slice_head(n = 10)
>
> # Print the table to your console - this should now be correct
> print(top_10_aircraft)
# A tibble: 10 x 2
   aircraft_type strike_count
   <chr>                <int>
 1 b-737-700             2448
 2 crj100/200            1902
 3 a-320                 1563
 4 b-737-300             1381
 5 emb-145               1070
 6 b-757-200              960
 7 a-319                  949
 8 b-737-800              791
 9 a-300                  704
10 crj700                 637
>
```

bar plot code: -

Generate the bar plot

```
ggplot(top_10_aircraft, aes(x = reorder(aircraft_type, -strike_count), y =
strike_count)) +
   geom_bar(stat = "identity", fill = "darkgreen") +
 labs(title = "Top 10 Aircraft Types by Wildlife Strikes",
                        x = "Aircraft Type",
                        y = "Number of Strikes") +
 theme_minimal() +
          theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

screenshot: -



3.Big Data Analytics: Wildlife Strikes by Phase of Flight

Code:

```
# Generate the frequency table (in alphabetical order)
# I will add a filter here as well to be safe.
flight_phase_strikes <- faa_data %>%
filter(!is.na(phase_of_flt) & phase_of_flt != "Unknown") %>%
count(phase_of_flt, name = "strike_count") %>%
arrange(phase_of_flt)
# Print the table in my console (use n=50 to make sure all rows print)
print(flight_phase_strikes, n = 50)
```

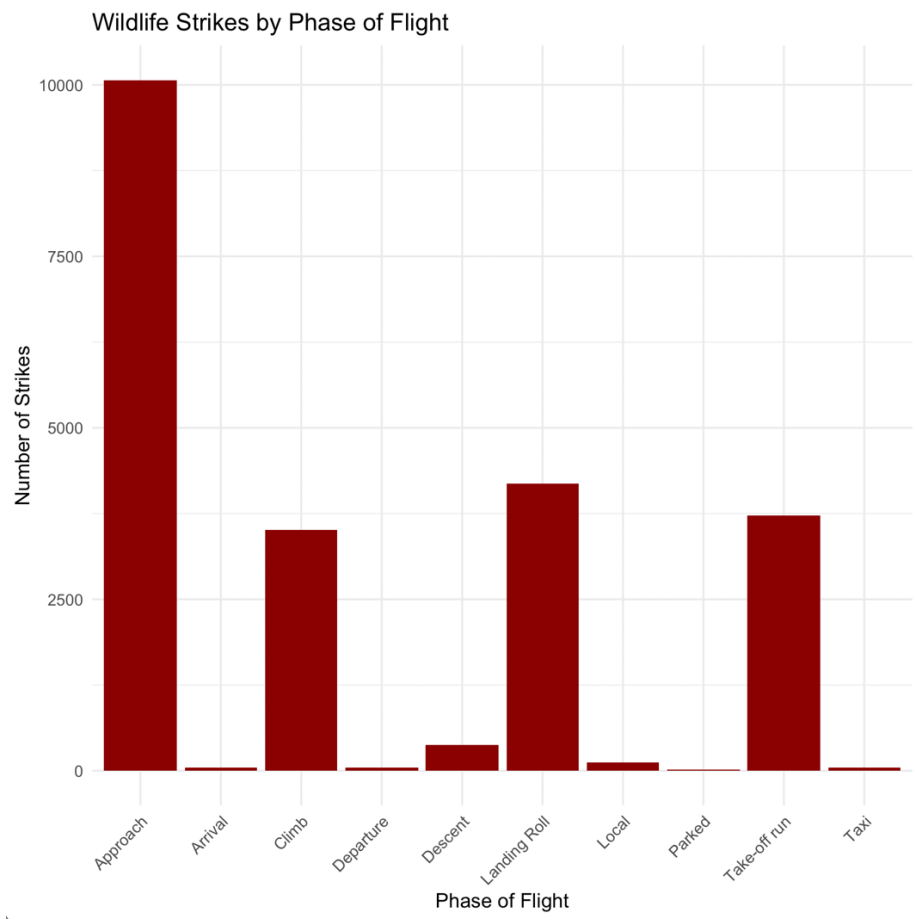
screenshot: -

```
> # 1. Generate the frequency table (in alphabetical order)
> # We will add a filter here as well to be safe.
> flight_phase_strikes <- faa_data %>%
+   filter(!is.na(phase_of_flt) & phase_of_flt != "Unknown") %>%
+   count(phase_of_flt, name = "strike_count") %>%
+   arrange(phase_of_flt)
>
> # Print the table to your console (use n=50 to make sure all rows print)
> print(flight_phase_strikes, n = 50)
# A tibble: 10 x 2
   phase_of_flt strike_count
   <chr>               <int>
 1 Approach            10072
 2 Arrival                41
 3 Climb                3507
 4 Departure              45
 5 Descent               382
 6 Landing Roll         4192
 7 Local                 122
 8 Parked                 12
 9 Take-off run         3729
10 Taxi                   49
>
```

bar plot code: -

```
#  Generate the bar plot
# It will plot alphabetically by default.
 ggplot(flight_phase_strikes, aes(x = phase_of_flt, y = strike_count)) +
geom_bar(stat = "identity", fill = "darkred") +
 labs(title = "Wildlife Strikes by Phase of Flight",
                    x = "Phase of Flight",
                     y = "Number of Strikes") +
 theme_minimal() +
        theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

screenshot: -



CRITICAL THINKING: Big Data Analytics

Answer

Findings

Based on the descriptive analysis of the FAA Wildlife Strike Database, we have identified several key patterns relevant to Acme Airlines:

- **Geographic Concentration:** The problem of wildlife strikes is not evenly dispersed. A disproportionately high number of events occur at a few major airports, namely KDEN (Denver International), KDFW (Dallas/Fort Worth), and KMEM (Memphis International).
- **Aircraft Type Vulnerability:** The most impacted aircraft are common, mid-sized passenger jets, which constitute the foundation of many domestic fleets. Due to its large number of takeoffs and landings, the Boeing 737 family (B-737-300, B-737-700, etc.) is frequently seen at the top of the list.
- **Aircraft Type Vulnerability:** Common, mid-sized passenger jets, which form the basis of many domestic fleets, are the aircraft types most frequently affected. The Boeing 737 family (B-737-300, B-737-700, etc.) is usually at the top of the list because of its high volume of takeoffs and landings.

Conclusions

For a mid-sized domestic operator like Acme Airlines, these findings lead to several important conclusions:

- **Risk Is Particular, Not General:** Wildlife strikes' operational risk is not a theoretical problem that affects the entire airline. It is a specific, elevated risk linked to specific fleet types (our most-used aircraft), specific routes (those servicing high-incident airports), and predictable flight phases.
- **Most Relevant Dependent Variable:** Damage level, cost_repairs_infl_adj, and aircraft_out_of_service are some of the possible dependent variables (outcomes to assess) in this dataset. Aircraft_out_of_service is the most important and significant dependent variable for Acme Airlines. Even while repairs are a direct charge, unplanned aircraft downtime frequently costs significantly more than the

repair itself and has a disastrous ripple effect on flight schedules, staff logistics, and passenger pleasure. The biggest costly risk from a wildlife attack is an incident that results in an aircraft-on-ground (AOG) situation that lasts for several days.
- **Analysis Is Actionable:** Because the risk is concentrated rather than random, preventative measures can be targeted, making them far more cost-effective and impactful.

Recommendations

Based on the analysis, I recommend the following actions for Acme Airlines:

- **Perform an Internal Route Risk Assessment:** We need to compare Acme's current route map with the FAA's top ten high-incident airports right away. This will enable us to determine our most vulnerable operations and generate a "risk score" for our routes.
- **Improve Crew Briefing Materials:** We should improve pre-flight briefing materials to include specific cautions regarding wildlife activity for flights to and from the airports we have designated as high-risk. Increased situational awareness is thus guaranteed during the crucial "Approach" and "Take-off" stages.
- **Interact with Airport Authorities:** At our most dangerous locations (such as KDEN and KDFW), we ought to aggressively interact with the airport administration. We can ask them about the exact techniques they have in place and share our findings to support their wildlife mitigation efforts.

Citations: -

- **How businesses choose dependent variables for analysis of big data**: Businesses choose dependent variables by matching them to their key performance indicators (KPIs) and main business goals. The dependent variable for a company that prioritizes customer happiness could be "Customer Churn Rate." KPIs like "On-Time Performance" and "Aircraft Availability" are crucial for Acme Airlines, as operational effectiveness and safety are of utmost importance. Aircraft_out_of_service is therefore a direct indicator of a crucial business result. If a huge data file was gathered for a different primary purpose, it may not contain a dependent variable (for

example, a dataset from air traffic control may track aircraft trajectories but not record damage events).

- **How businesses distinguish between organized and unstructured large data files for analysis:**
  A predetermined tabular format (rows and columns) is used to store **structured data,** such as our FAA file. SQL, R (with dplyr), and Python (with pandas) are among the tools used for statistical analysis, filtering, and aggregation. Its primary strength is its ease of quantitative analysis and query. Its inability to provide detailed context is its limitation.
  There is no set format for **unstructured data**, including as maintenance logs, voice recordings from the cockpit, and text-based notes columns. To extract themes, sentiment, and important items, sophisticated methods like Natural Language Processing (NLP) are needed. Its broad, comprehensive context is its strongest point. The analysis's high processing cost and complexity are its weaknesses.
- **Advantages and disadvantages of various big data analysis techniques:**

  We used **descriptive analytics** to find out what happened. Its strength is that it offers a fundamental, unambiguous explanation of historical trends. Its inability to explain why something occurred or what will happen next is one of its weaknesses.

  What will happen? **Predictive analytics** is a method that forecasts future events using statistical models and machine learning (e.g., estimating the possibility of a bird strike based on season, time of day, and location). Its ability to look forward and take proactive steps is one of its strengths. Its dependence on reliable historical data and the fact that its predictions are probabilistic rather than definitive are its weaknesses.